

SOVETSKAYA GAVAN, Russian Federation

WMO: 317700

Lat: 49.000N

Lon: 140.300E

Elev: 24

StdP: 101.04

Time Zone: 10.00 (E10)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.6	-23.7	-31.5	0.2	-24.2	-29.9	0.2	-22.3	14.1	-9.1	12.3	-8.8	1.8	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	26.0	18.9	23.9	18.1	22.0	17.1	20.1	24.4	18.9	22.5	17.9	20.9	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.3	21.9	17.5	12.6	20.7	16.7	11.9	19.5	57.8	24.8	54.0	22.6	51.0	21.0	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.0	8.2	-29.0	30.4	2.1	1.4	-30.5	31.4	-31.7	32.2	-32.9	33.0	-34.4	34.0		
(5)			-29.2	22.5	2.0	0.9	-30.7	23.1	-31.9	23.7	-33.0	24.2	-34.5	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-14.8	-13.1	-6.0	1.3	7.1	11.6	16.0	17.6	13.6	5.9	-4.1	-12.0
	DBStd	11.90	4.56	4.72	4.24	3.40	3.31	3.31	2.81	2.34	3.33	3.54	5.51	4.80
	HDD10.0	3539	769	648	496	262	103	18	0	0	7	131	423	683
	HDD18.3	5995	1028	881	754	511	349	203	83	42	145	384	673	942
	CDD10.0	620	0	0	0	1	12	66	185	236	116	6	0	0
	CDD18.3	34	0	0	0	0	0	2	10	19	3	0	0	0
	CDH23.3	228	0	0	0	0	4	29	72	96	28	0	0	0
	CDH26.7	32	0	0	0	0	0	4	14	11	3	0	0	0
Wind	WSAvg	3.0	3.3	2.8	3.1	3.1	2.6	2.1	2.1	2.2	2.7	3.6	3.9	4.1
Precipitation	PrecAvg	739	15	21	41	54	70	70	77	106	114	93	45	36
	PrecMax	1292	43	81	116	158	138	194	231	535	339	226	138	164
	PrecMin	547	0	0	4	9	15	8	2	13	0	5	1	1
	PrecStd	166	12	21	25	31	34	43	51	87	72	56	36	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.0	1.3	7.6	18.2	22.8	27.1	28.7	28.1	26.5	18.4	10.5	0.6
		MCWB	-1.5	-2.6	2.3	8.2	12.1	17.8	21.2	20.5	18.5	10.8	5.2	-0.3
	2%	DB	-3.5	-2.0	4.3	12.7	18.8	23.0	25.3	25.8	23.4	15.7	7.4	-1.1
		MCWB	-4.9	-4.2	0.4	5.9	11.1	16.3	19.4	19.3	16.9	10.0	3.4	-2.3
	5%	DB	-5.3	-3.8	2.1	9.1	16.0	20.3	22.6	24.0	21.1	13.4	5.2	-2.8
		MCWB	-6.9	-5.5	-0.6	3.9	9.3	15.0	18.2	18.7	15.8	9.2	2.9	-4.0
	10%	DB	-7.3	-5.4	0.5	6.6	13.1	17.7	20.6	22.1	19.0	11.7	3.8	-4.3
		MCWB	-8.8	-6.9	-1.6	2.8	8.4	13.5	17.2	18.0	14.7	8.0	1.4	-5.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.6	-1.9	2.6	8.7	13.7	19.1	21.7	21.7	19.2	13.4	5.8	0.0
		MCDB	-1.1	0.4	6.5	16.6	21.2	25.1	28.0	25.6	24.7	15.4	8.8	0.5
	2%	WB	-4.8	-3.8	0.9	6.6	11.5	16.9	20.0	20.4	17.9	11.2	4.3	-2.1
		MCDB	-3.6	-2.2	3.1	11.7	17.2	22.7	24.6	24.3	21.6	13.9	6.5	-1.3
	5%	WB	-6.8	-5.3	-0.3	4.3	10.0	15.3	18.5	19.3	16.8	10.0	3.2	-3.8
		MCDB	-5.4	-3.9	1.7	8.3	14.6	19.4	21.9	22.7	19.6	12.2	5.1	-2.7
	10%	WB	-8.8	-6.9	-1.4	2.8	8.7	13.8	17.4	18.4	15.7	8.6	1.7	-5.7
		MCDB	-7.3	-5.4	0.4	5.9	12.5	17.2	20.1	21.3	18.1	11.3	4.0	-4.5
Mean Daily Temperature Range	5% DB	MDBR	8.3	10.2	9.1	7.7	8.4	8.2	7.5	7.3	9.0	8.5	7.5	7.7
		MCDBR	6.8	10.9	10.3	13.6	15.1	14.1	12.5	11.6	12.2	12.0	8.9	7.6
		MCWBR	6.1	9.0	7.5	7.8	7.9	7.8	7.2	6.0	6.6	7.7	6.7	7.1
	5% WB	MCDBR	5.6	9.7	9.4	12.7	12.9	12.9	11.5	9.9	9.9	10.2	8.0	7.0
		MCWBR	5.2	8.2	7.2	7.5	7.4	7.6	6.9	5.7	6.0	7.3	6.4	6.8
Clear-Sky Solar Irradiance	taub	0.289	0.323	0.394	0.486	0.532	0.505	0.473	0.428	0.387	0.393	0.347	0.301	
	taud	2.145	2.031	1.902	1.828	1.845	2.022	2.178	2.324	2.365	2.165	2.147	2.134	
	Ebn at Noon	744	797	791	757	741	768	788	811	809	711	655	664	
	Edh at Noon	82	117	159	192	197	166	140	116	101	103	83	73	
All-Sky Solar Radiation	RadAvg	1.35	2.46	3.67	4.40	4.79	5.09	4.66	4.25	3.64	2.37	1.39	1.01	
	RadStd	0.14	0.19	0.29	0.49	0.39	0.54	0.46	0.39	0.34	0.15	0.12	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	+0.82	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page